

Pranav Ponnivalavan

9-6-5, Higashi Kasai, Edogawa-Ku, Tokyo, Japan | 070-2024-9839 | pranavponni2003@gmail.com

I work on multi-finger manipulation research at Waseda University under *Tetsuya Ogata & Shigeki Sugano* & currently also do cognitive HRI research with *Yudai Tanaka, Karthik Mahadevan* and *Justin W. Hart* remotely and will be visiting (UT) University of Texas at Austin from Jan-Apr 2027.
Seeking PhD opportunities, R&D positions.

EDUCATION

Waseda University, Tokyo, Japan Master of Engineering, Modern Mechanical Engineering (Robotics Research)	<i>Expected Sep 2027</i>
Waseda University, Tokyo, Japan Bachelor of Engineering, Modern Mechanical Engineering (Robotics Research)	<i>Sep 2021 – Sep 2025</i>
APL Global School, Chennai, India CIE AS & A Levels, IGCSE, G3 – G12	<i>Apr 2011 – Mar 2021</i>
K International School, Tokyo, Japan K1 – G2	<i>Jul 2007 – Mar 2011</i>

RESEARCH PUBLICATIONS

- M-TaSA: Learning Multi-Finger Self-Touch for Sensory Attenuation in Dexterous Manipulation
Pranav Ponnivalavan*, Satoshi Funabashi, Alexander Schmitz, Tetsuya Ogata, Shigeki Sugano
IEEE Transactions on Robotics (T-RO) - **Under Review*
- TaSA: Two-Phased Deep Predictive Learning of Tactile Sensory Attenuation for Improving In-Grasp Manipulation
Pranav Ponnivalavan*, Satoshi Funabashi, Alexander Schmitz, Tetsuya Ogata, Shigeki Sugano
IEEE ICRA 2026 · International Conference on Robotics and Automation - **Under Review*
- PACT: Posture-Aligned Co-Design Technique for Hand Configuration and Retargeting in Multi-Finger Imitation Learning
Koyo Sasazaki, Satoshi Funabashi, **Pranav Ponnivalavan**, Alexander Schmitz, Tetsuya Ogata, Shigeki Sugano
IEEE ICRA 2027 · International Conference on Robotics and Automation - **Under Review*
- A Dual-Surface Tactile Fingertip for Contact-Rich Dexterous Manipulation of a Multi-Fingered Hand
Steven Oh, Satoshi Funabashi, Hiroki Niimi, Tai Yamada, Kazutaka Omori, Pranav Ponnivalavan, Tetsuya Ogata, Shigeki Sugano
IEEE ICRA 2027 · International Conference on Robotics and Automation
- Depth Conditioning for Bimanual Cable Routing: Gains Are Confined to Phases That Constrain the Depth Axis
Takeru Suzuki, Satoshi Funabashi, Haoyu Zhao, **Pranav Ponnivalavan**, Pei-Chun Chien, Shigeki Sugano
IEEE ICRA 2027 · International Conference on Robotics and Automation - **Under Review*

SKILLS

Programming: Python, C, C++(Arduino), Bash
Languages: English (Native), Japanese (Intermediate – N3), Tamil (Native)
Software: ROS, Rviz, Gazebo, Autodesk Fusion, OnShape, KiCAD
Robots & Tele-op: Universal Robots(3e, 5e), Allegro Hand (v3, v4), XELA uSkin sensors, Tesollo DG5F, DoBot Bimanual Arms, FingerVision sensors, GelSight sensors, MANUS metagloves, Robotiq Gripper

WORK EXPERIENCE

Waseda University R&D Center <i>Undergraduate + Graduate Researcher</i>	Shinjuku-Ku, Tokyo <i>Mar 2024 – Present</i>
- Research on sensory attenuation for tactile-driven multi-finger in-grasp manipulation using Allegro Hand and uSkin tactile sensors. - Currently under review in <i>IEEE Transactions on Robotics</i>	

- Accepted conference paper at ICRA *IEEE: International Conference on Robotics & Automation* 2026.
- Co-worked with 2 PhD students, 2 master students (M1 & M2), assisted and managed 7 bachelor students: (2 B4 students & 5 B3 bachelor students)

FingerVision

Koto-Ku, Tokyo

Robotic Process & AI Engineer

Oct 2025 - Present

- Built an end-to-end robotics pipeline (data collection → training → inference) and validated usability of the robot arm setup.
- Created datasets and evaluated inference across multiple tasks (zipper, box pick inserts, USB plug insertion, yellow work picking, cable manipulation, cutting vegetables).
- Worked with many clients: Pokémon, Ajinomoto, Mitsubishi Electric, Asahi Electric, etc.
- Iterated a 3D-printed fixture for FingerVision tactile sensing to improve stability and repeatability; debugged training failures (OOM, data loading, index/array bugs).

O-ID

Shibuya-Ku, Tokyo

Humanoids Design Engineer

Sep 2025 – Feb 2026

- Designed a bilateral, stable, transportable base architecture with secure dual-arm mounting and ergonomic tabletop reach.
- Engineered electronics integration (internal bays, thermal/ventilation strategy, robust component mounting, and cable-routing paths).
- Delivered a fully parametric CAD + manufacturing package (assembly, 2D drawings, BOM, STEP handoff) and final build documentation with validation under specified load cases.

TCS – Tata Consultancy Services Research

Minato-Ku, Tokyo

Robotics & Digital Engineering Intern

Jun 2024 – Aug 2024

- Designed and 3D-printed jigs in Fusion 360 for a Universal Robot 5e; extracted force data using a jig attached to the robot setup.
- Built and evaluated ML models to classify swipe motions as success/failure (Logistic Regression, Random Forest, Gradient Boosting, RNN).

Daimler Trucks Asia

Kawasaki, Kanagawa

Factory Automation Intern

Feb 2023 – May 2023

- Supported analysis of robot equipment maintenance and utility operation data to identify optimization opportunities.
- Coordinated stakeholders and supported KPI tracking, budgeting/forecasting, and operational metrics for facilities initiatives.

EXTRACURRICULAR EXPERIENCE

GDSC – Google Developers Student Club

Shinjuku-Ku, Tokyo

Member of Education and Finance Team

Sep 2022 – Apr 2024

- Monitored and documented financial statements and budgeting from facilitators and students for the finance team
- Conducted seminars and workshops on latest software developments on AI and coding for freshmen and sophomores.

TA – Teaching Assistant

Shinjuku-Ku, Tokyo

Mechanical Engineering Laboratory Basic

Sep 2024 – Jan 2025

- Teaching Assistant for second-year course that dealt with hands-on lab and building of robots from scratch
- Helped students with doubts and questions of code, software, hardware and was also in charge of discussion handling.